

ELECTIVECOURSE-4.5

UME752: NOISE, VIBRATION AND HARSHNESS				
	L	T	P	Cr
	3	0	0	3.0
Course Objectives: To impart knowledge to the students about the fundamentals of noise, vibration and harshness and their impact in real life applications. To enable the students to identify the role of NVH engineers in determining the quality of noise and vibration in development stages of a new vehicle. To impart knowledge on advanced NVH control technologies.				
Fundamentals of Automotive NVH: Basics of NVH study- and terminologies- Narrow band and octave band types (Example Topics), NVH Sources - Vibration & Noise, Structure-Borne & Air-borne NVH differences, Basics of order of vibration- frequency domain function- FFT features- ISO standards- Whole-Body Vibration analysis- Human sensitivity and weighting factors related to NVH, Harshness				
Automotive NVH performance development: Structure Borne NVH - Booming Noise, Steering Vibration/shake/shimmy, ABC pedals vibration, Seat vibration, Mirror vibrations, Road induced NVH, Airborne NVH - Powertrain Radiated noise, Intake / Exhaust noise, Aerodynamics noise, Tire Noise, Operational conditions - Idle NVH, Constant Speed / Acceleration NVH, Exterior Noise & Regulations - Pass by Noise, Tail pipe noise, Engine radiated Noise, Standards (Govt & Automotive Agencies),EV NVH - motor/generator whine				
Basics of Structure Borne & Air Borne NVH: Structure-borne NVH - Resonance concept, Vibration transmissibility, Stiffness / Isolation / Damping concepts, Airborne NVH - Absorber / Barrier concepts, Transmission Loss, Insertion loss, Absorption coefficient				
NVH Test Facilities and Instrumentation: Basic Instrumentation – Transducers, Accelerometer, Impact Hammer, Microphones, Anechoic/Semi-Anechoic rooms, Reverberation chambers, Alpha cabin, Kundts tube				
Advanced NVH development: LASD damping, Quiet Steel, Active Noise Cancellation, Active Sound Enhancement, Semi-active and active mounts, Hydra mounts, AVAS - Acoustic Vehicle Alert System				
CAE-NVH development with Hands-On: Basic FRF analysis, Basic Transmission Loss analysis				
Course Learning Outcomes (CLOs) After completion of this course, students will be able to:				

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1. Apply the principles of sound and vibration to solve related real-life problems, especially in the automotive sector.
2. Identify and analyse the sources of automotive noise and vibration and apply the knowledge of an NVH engineer for noise and vibration control in a new vehicle development program.
3. Critically evaluate and use the various sound and vibration measurement techniques for automotive NVH. Compute sampling, statistical, and frequency analysis of various data obtained from sound and vibration measurements.
4. Apply the CAE tools for noise and vibration simulation, in particular the modal and frequency response analysis.

Text Book(s)

1. Anton Fuchs, Eugenius Nijman, Hans-Herwig Pribsch, Automotive NVH Technology, Springer, 2016.
2. James E. Duffy, Modern Automotive Technology, Goodheart-Willcox Pub, 2013.
3. Frank J. Fahy, Foundations of Engineering Acoustics, Academic Press, Elsevier, 2000.
4. Lewis H. Bell, Industrial Noise Control, Fundamentals and Applications, Marcel Dekker, 1994.

Reference Books

1. M. L. Munjal, Noise and Vibration Control, World Scientific Press: Singapore, 2014.
2. Trelleborg Vibracoustics, Automotive Vibration Control Technology Fundamentals, Materials, Construction, Simulation, and Applications. Vogel Communications Group GmbH & Company KG, 2015.

Evaluation Scheme

Sr. No.	Evaluation elements	Weightage (%)
1	MST	25
2	EST	45
3	Sessional (May include the following) Assignment, Sessional (Assessment: Quizzes, Project (Including report, presentation etc.)	30

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